

REPORT ON MINI PROJECT

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TITLE: AMAZON PRODUCT SALES DATA

PROJECT DOMAIN: SALES

TIMELINE: 2025

SUBMISSION DATE: 08.01.2026

MENTOR NAME: KUMARAN M

DATA SET LINK:

Raw Data Set:

<https://drive.google.com/file/d/1EEghPZkfb8dagpFcDSMuGIGQxjgTvTbd/view?usp=sharing>

Cleaned data set:

https://docs.google.com/spreadsheets/d/1mZF3tYI3IMWy1PJv2T_5zqmUgWoG4A5h/edit?usp=sharing&oid=111532347864707315771&rtpof=true&sd=true

Amazon Product Sales Data – Cleaning, Transforming & Visualization

This project aims to analyse and visualize the data from amazon e-commerce product performance by evaluating pricing, discounts, customer engagement, and marketing effectiveness across categories to support data-driven business decisions. The dataset contains product title, product rating, total reviews, purchased last month, discount price, original price, best seller, sponsored, coupon, delivery date, product image URL, product page URL, product category, discount percentage, data collected date, data collected time.

Objectives:

- To Analyze product performance and customer demand
- To Evaluate pricing and discount strategies
- To Measure customer satisfaction using ratings and reviews
- To Assess marketing impact of sponsored products and coupons
- To Compare category-level performance
- To Monitor delivery and data collection

Tools and techniques used:

Microsoft Excel - Data Cleaning, Transformation and Calculations.

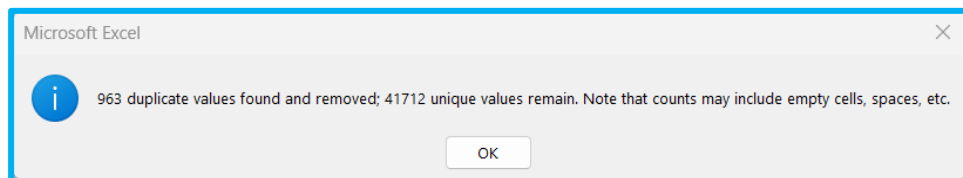
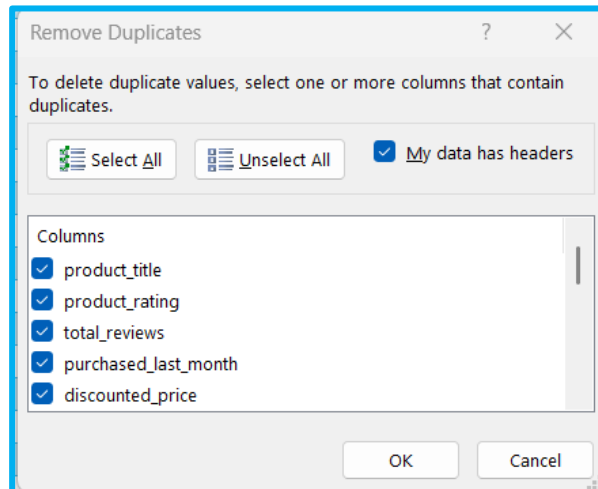
Power BI- Interactive Dashboard and Visual analytics.

Data Visualization Techniques – Bar chart, Line chart, column chart, Tree chart, donut chart, KPI, Gauge, Slicer.

DATA CLEANING

1.Remove Duplicates

I started to clean the data by removing duplicates.



2. Missing Values in Original Price:

=IF (ISBLANK (F2045) , AVERAGE (\$F\$2 : \$F\$41713) , F2045)

[illegible][illegible]

3. Dilimited Columns

I have Dlimited the columns into date and time

24-08-2025 21:45	24 August 2025	21:45:00
24-08-2025 21:51	24 August 2025	21:51:00
21-08-2025 11:44	21 August 2025	11:44:00
24-08-2025 21:42	24 August 2025	21:42:00
24-08-2025 21:48	24 August 2025	21:42:00
24-08-2025 21:42	24 August 2025	21:48:00
24-08-2025 21:43	24 August 2025	21:43:00
21-08-2025 11:52	21 August 2025	11:52:00
21-08-2025 12:13	21 August 2025	12:13:00
24-08-2025 21:47	24 August 2025	21:47:00
24-08-2025 22:00	24 August 2025	22:00:00
21-08-2025 12:14	21 August 2025	12:14:00
24-08-2025 21:52	24 August 2025	21:58:00
24-08-2025 21:58	24 August 2025	21:52:00

4. Added New Column:

I have added a new column Discount Percentage.

Discount percentage
0%
0%
0%
22%
0%
0%
0%
14%
16%
5%
15%
6%
6%
6%

TRANSFORMATION

1. Transformation of Data

I have transformed the data from excel file to power bi

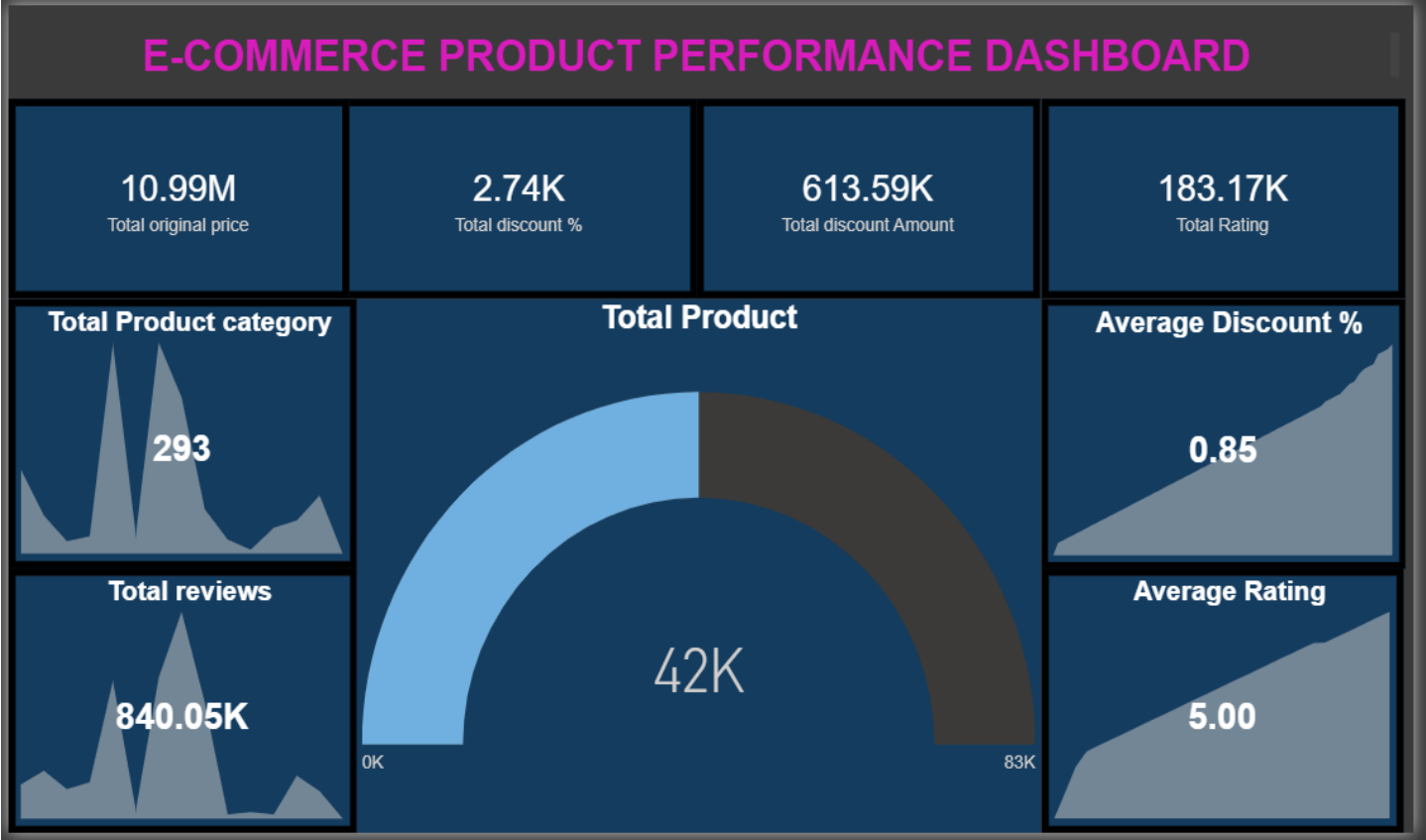
[illegible]

2. Calculating Measures

I have calculated some measures are Product Rating, Total Reviews, Purchased Last Month, Discount Price, Original Price, Discount Percentage and added some columns by using Dax.

[illegible]

DASHBOARD



1. Executive Overview

This dashboard provides a **high-level summary of product pricing, discounts, ratings, and customer engagement**. It is designed for **quick executive decision-making** with KPIs prominently displayed at the top.

2. KPI Card

- **Total Original Price (10.99M):** Indicates the total listed value of all products.
- **Total Discount Percentage (2.74K):** Represents cumulative discount intensity across products.
- **Total Discount Amount (613.59K):** Shows total monetary value given as discounts.
- **Total Product Rating (183.17K):** Reflects overall customer feedback volume.

3. Product Volume Indicator (Centre Gauge)

- The **gauge visual shows Total Products = 42K**, with a scale up to 83K.
- This helps track **catalogue size and growth capacity** at a glance.

4. Category Coverage

- **Total Product Category = 293**
- Indicates a **wide and diversified product portfolio**, reducing dependency on a single category.

5. Customer Engagement

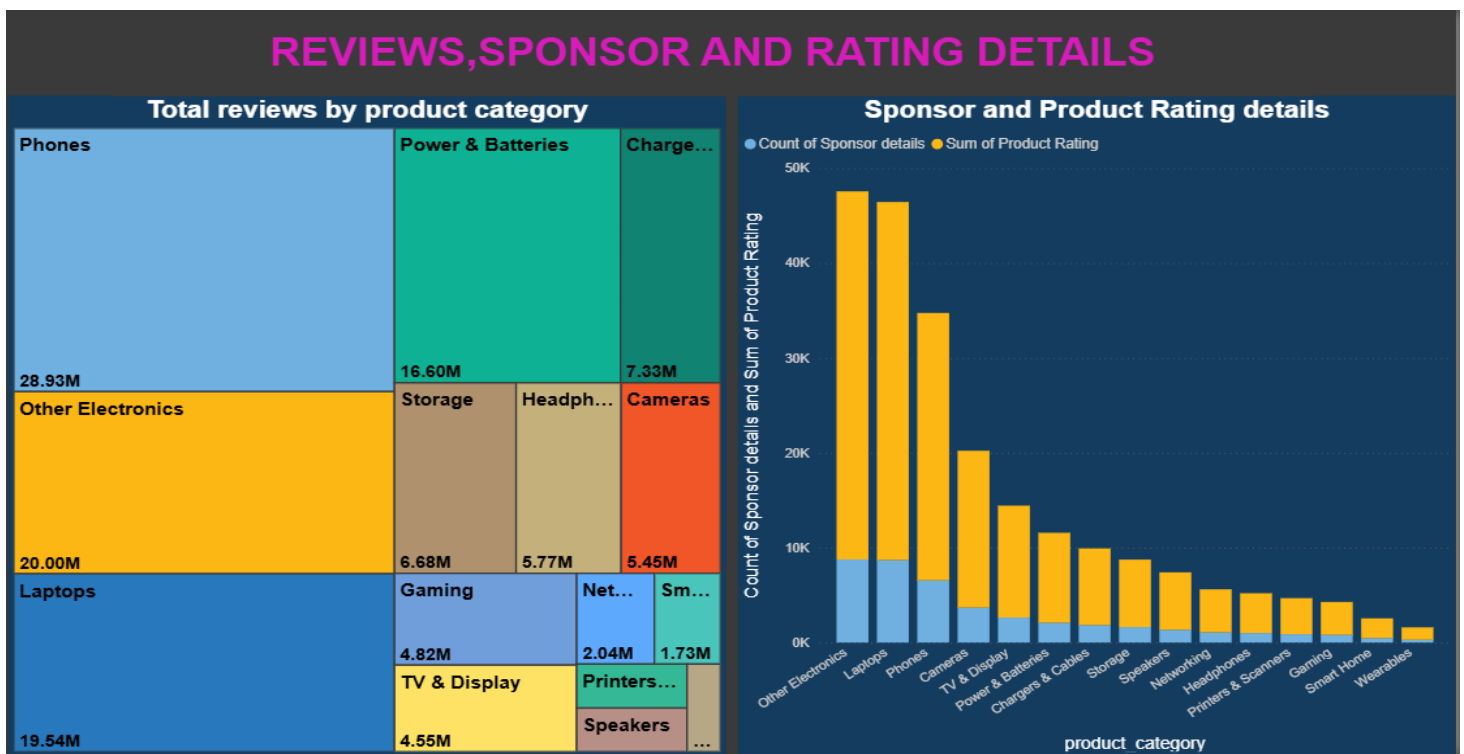
- **Total Reviews = 840.05K**
- Suggests **strong customer interaction and marketplace activity**.
- High review volume improves product credibility and ranking.

6. Pricing & Discount Effectiveness

- **Average Discount % = 0.85**
- Shows the **overall discount level offered to customers**.
- Useful for evaluating whether pricing is aggressive or conservative.

7. Product Quality Insight

- **Average Rating = 5.00**
- Indicates **excellent customer satisfaction** across products.
- Strong signal for brand trust and repeat purchases.

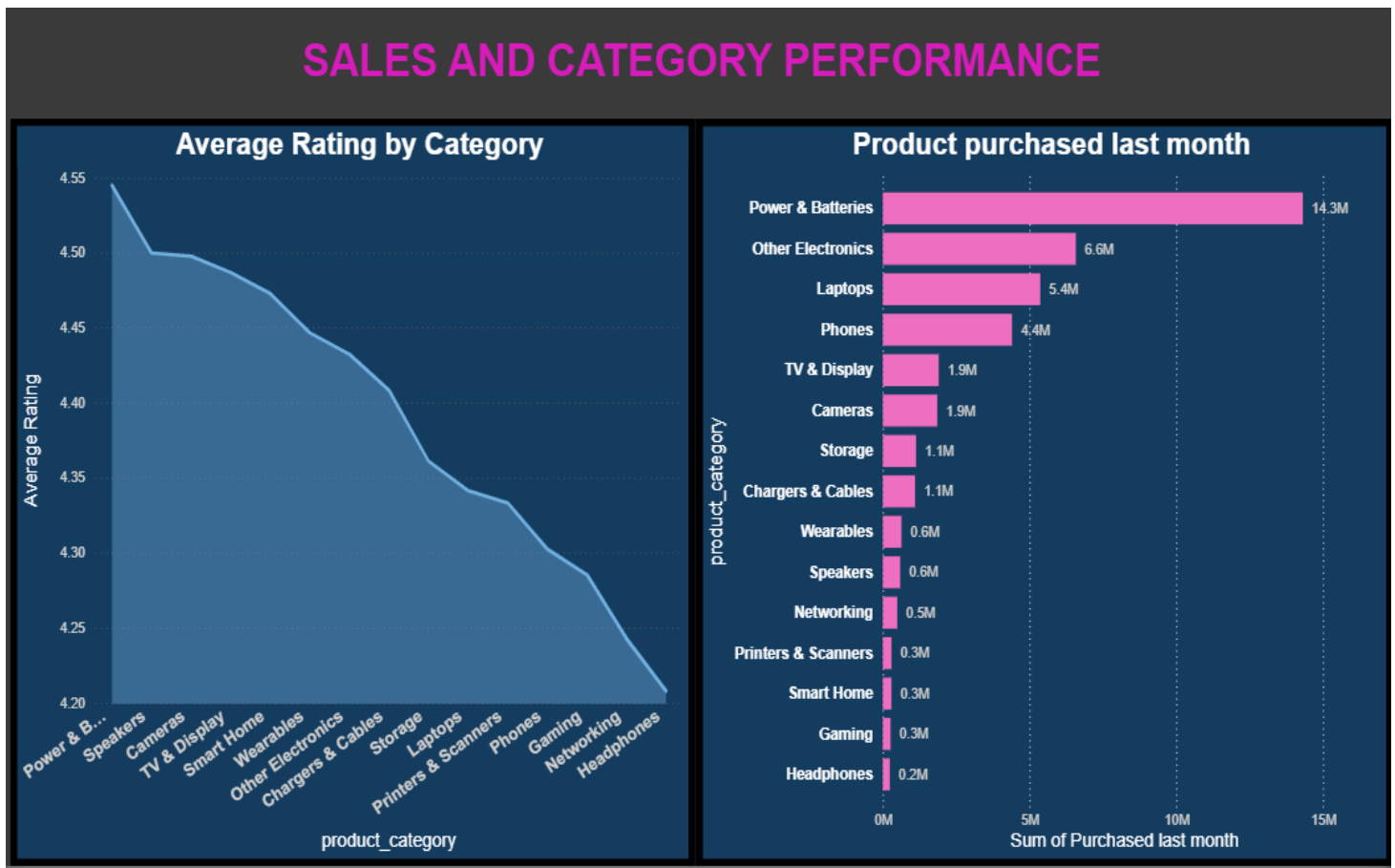


Total Reviews by Product Category (Tree map)

- The tree map shows the **distribution of total customer reviews across product categories**.
- **Phones (28.93M reviews)** and **Other Electronics (20.00M reviews)** dominate customer engagement.
- **Laptops (19.54M reviews)** and **Power & Batteries (16.60M reviews)** also show strong interaction.
- Categories such as **Gaming, TV & Display, Cameras, and Networking** have comparatively lower review volumes.

Sponsor and Product Rating Details (Clustered Column Chart)

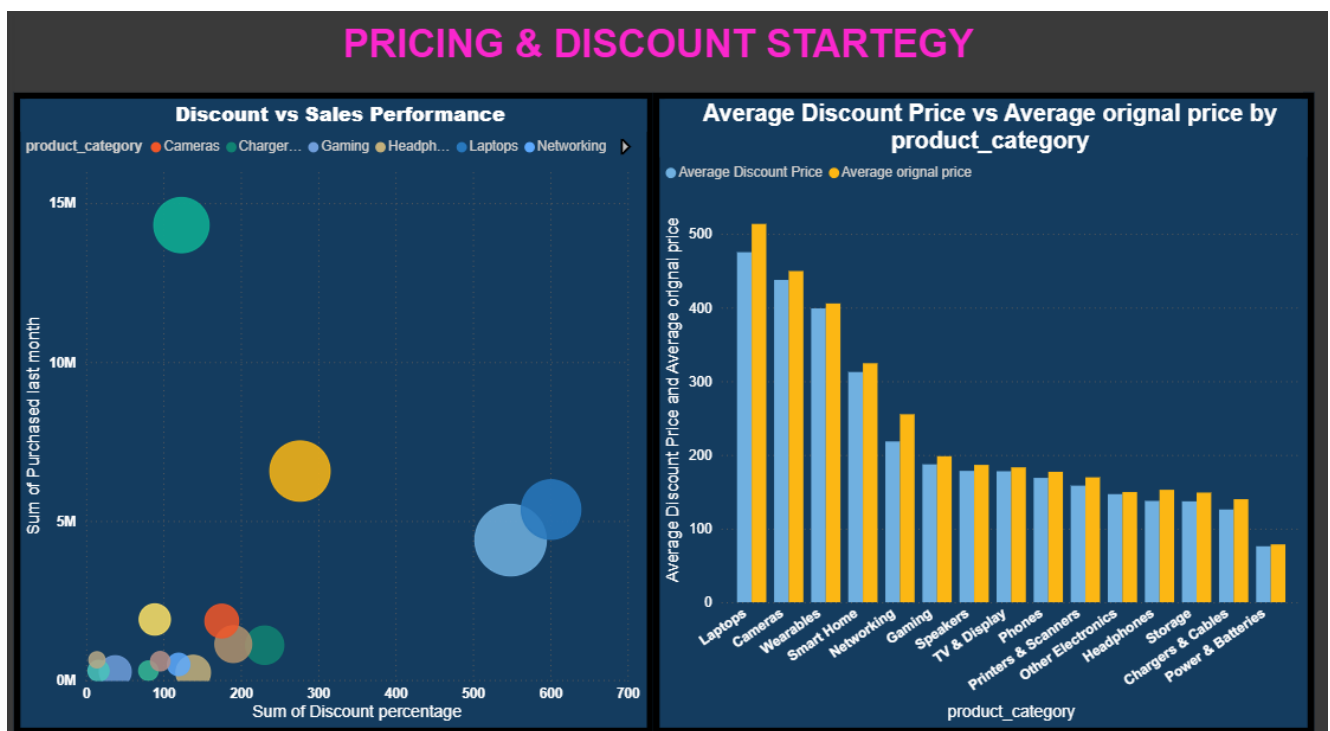
- This chart compares **count of sponsored products** with the **sum of product ratings** across categories.
- Categories like **Other Electronics, Laptops, and Phones** have:
 - High product ratings
 - Higher presence of sponsored products



Sales and Category Performance

- The dashboard visualizes **category-wise customer satisfaction and sales performance** using average product ratings and purchase volume from the previous month.
- The **Average Rating by Category** chart shows that **Power & Batteries, Speakers, and Cameras** receive the highest customer ratings, indicating strong product quality and positive user experience.
- Categories such as **Networking, Gaming, and Headphones** record comparatively lower average ratings, suggesting the need for product quality improvements or better customer support.
- The **Product Purchased Last Month** chart highlights **Power & Batteries** as the top-selling category with the highest purchase volume.
- **Other Electronics** and **Laptops** also demonstrate strong sales performance, contributing significantly to overall sales.

- Several categories, including **Speakers**, **Wearables**, and **Cameras**, exhibit good customer ratings but moderate sales, indicating potential opportunities for increased promotion and visibility.
- Categories such as **Smart Home**, **Gaming**, and **Headphones** show low sales volumes, reflecting niche demand or limited market penetration.
- The comparison of ratings and sales reveals that **high customer satisfaction does not always translate directly into high sales**, emphasizing the impact of pricing, necessity, and promotional strategies.
- Overall, this analysis supports strategic decisions related to **category prioritization, marketing focus, and inventory planning**.



Pricing & Discount Strategy

- The dashboard evaluates the relationship between **discount levels, sales performance, and pricing structure** across product categories.
- The **Discount vs Sales Performance** scatter plot shows that categories with **moderate discounts** often achieve **higher sales volumes** compared to categories offering very high discounts.
- **Power & Batteries** stands out with the **highest sales volume**, despite having a relatively lower total discount percentage, indicating strong demand driven by necessity rather than discounts.
- Categories such as **Laptops** and **Cameras** exhibit **higher discount percentages** along with moderate sales, suggesting that discounts are used to stimulate demand in higher-priced segments.
- **Gaming** and **Headphones** show lower sales even with noticeable discounts, implying limited discount effectiveness in these categories.

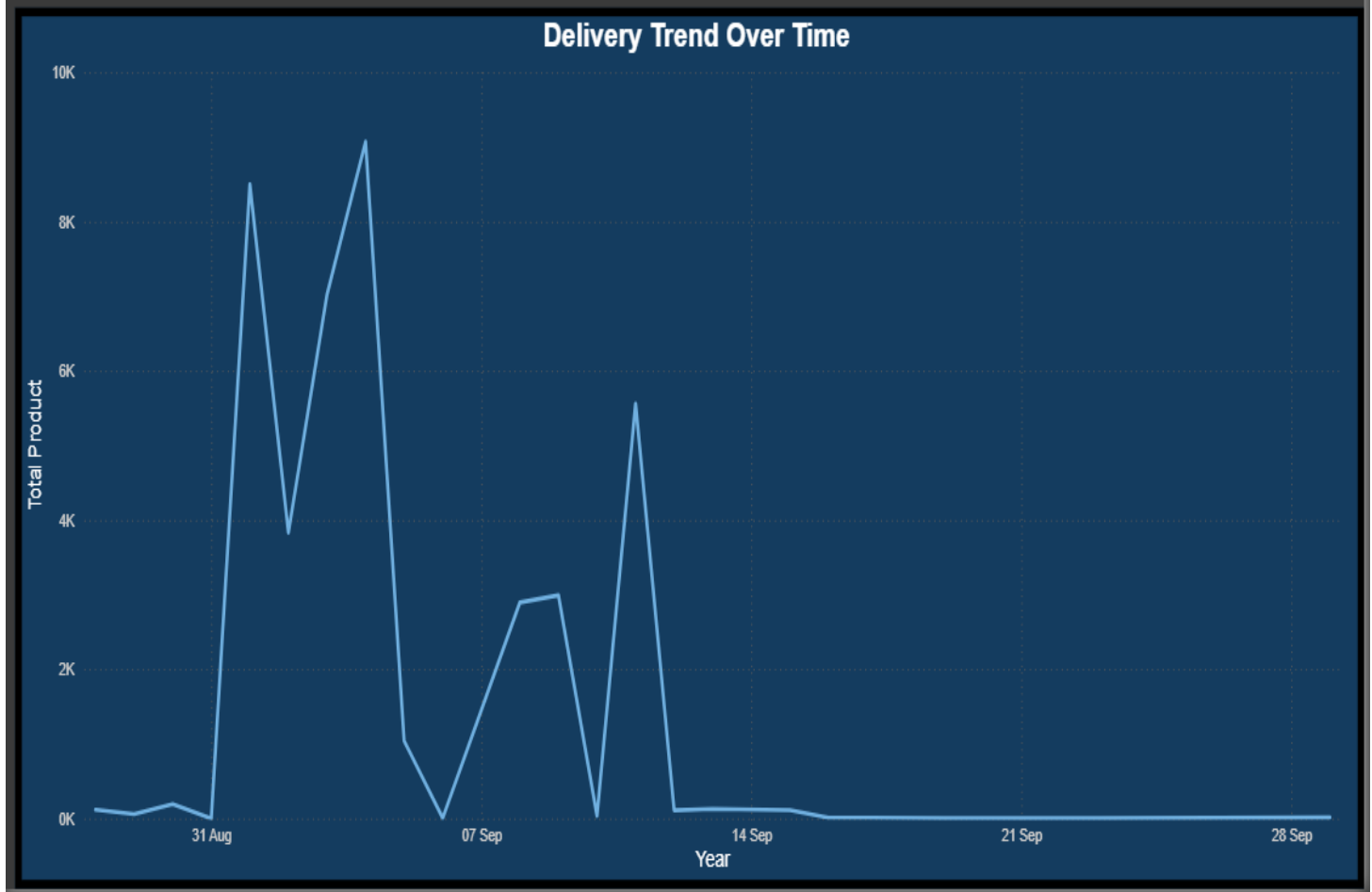
- The **Average Discount Price vs Average Original Price** chart reveals a consistent price gap across all categories, highlighting uniform discounting strategies.
- **Laptops and Cameras** have the highest original and discounted prices, reflecting their premium positioning in the product portfolio.
- **Power & Batteries** and **Chargers & Cables** are positioned at the lower end of the price spectrum, making them more accessible to price-sensitive customers.
- The analysis indicates that **discounts alone do not guarantee higher sales**, and product demand is also influenced by category necessity, pricing level, and customer preference.
- Overall, the visualization supports informed decisions on **pricing optimization, discount planning, and category-specific promotional strategies**.



MARKETING EFFECTIVENESS – DETAILED INSIGHTS

- The dashboard demonstrates that **marketing labels, sponsorships, and coupons directly influence customer purchasing behaviour**.
- **Best Seller status is strongly linked with effective discount messaging** rather than excessive price reductions.
- Sponsored promotions are most effective in **high-value and high-competition categories**.
- Coupon strategies are particularly impactful for **boosting immediate sales performance**.
- Overall, a **balanced marketing approach** combining organic visibility, moderate discounts, and targeted sponsorships yields optimal results.

OPERATIONAL INSIGHTS



Operational Insights: Delivery Trend Over Time

- The line chart illustrates the **trend of total products delivered over time**, covering the period from **late August to late September**.
- The **X-axis** represents the time period (dates), while the **Y-axis** shows the **total number of products delivered**.
- This visualization helps assess **delivery volume fluctuations**, operational consistency, and potential supply chain bottlenecks.

PRODUCT DETAIL

Sum of Product Rating	Sum of Discounted price	Best seller	Sum of Discount percentage	Sponsored	Sum of Purchased last month	product_category
33,787.90	38,10,600.95	No Badge	510.52	Organic	39,44,800.00	Laptops
20,995.77	9,63,540.23	No Badge	352.83	Organic	22,92,370.00	Phones
36,767.22	11,96,869.72	No Badge	242.19	Organic	54,86,400.00	Other Electronics
6,467.29	1,50,355.25	No Badge	182.50	Organic	7,86,505.00	Chargers & Cables
13,826.19	13,95,416.21	No Badge	147.32	Organic	13,49,885.00	Cameras
5,499.22	1,02,364.40	No Badge	141.23	Sponsored	13,80,995.00	Phones
2,072.80	49,226.79	No Badge	117.10	Sponsored	4,42,965.00	Storage
3,484.40	1,40,389.34	No Badge	116.79	Organic	2,50,090.00	Networking
2,862.29	95,116.26	No Badge	92.15	Organic	1,80,030.00	Headphones
4,875.60	1,94,192.16	No Badge	89.76	Organic	3,65,220.00	Speakers
3,667.10	1,37,335.32	No Badge	76.38	Organic	1,74,595.00	Printers & Scanners
6,129.29	3,06,318.19	No Badge	72.20	Organic	6,20,625.00	TV & Display
4,831.10	1,69,562.59	No Badge	67.81	Organic	6,32,075.00	Storage
5,488.40	1,19,455.98	No Badge	49.90	Organic	13,57,565.00	Power & Batteries
2,919.48	31,937.03	No Badge	43.16	Sponsored	92,04,815.00	Power & Batteries
2,435.29	1,37,054.31	No Badge	42.20	Sponsored	6,72,290.00	Laptops
537.50	17,493.75	Save 30%	37.50	Organic	12,500.00	Headphones
3,369.86	1,46,186.18	No Badge	36.09	Organic	2,41,065.00	Gaming
1,050.39	24,380.45	No Badge	29.14	Sponsored	1,70,960.00	Chargers & Cables
294.96	7,650.83	Limited time deal	21.58	Organic	4,750.00	Laptops
2,339.97	1,92,958.56	No Badge	20.83	Sponsored	4,19,895.00	Cameras
614.40	2,258.12	Limited time deal	19.66	Sponsored	16,86,000.00	Power & Batteries
2,023.10	1,41,639.12	No Badge	14.86	Organic	2,69,020.00	Smart Home
1,83,173.17	1,03,74,851.98		2,736.74		3,94,10,645.00	

The **Product Detail** table provides a consolidated view of product performance by combining **pricing, discounting, sales volume, promotion type, and category-level metrics**. It enables stakeholders to analyse how different product categories perform under various promotional and pricing strategies.



This dashboard focuses on **operational performance monitoring**, specifically analysing:

1. **Product delivery trends over time**
2. **Patterns in product data collection activity**

By comparing these two timelines, the dashboard helps assess **logistics efficiency, data consistency, and operational stability**.